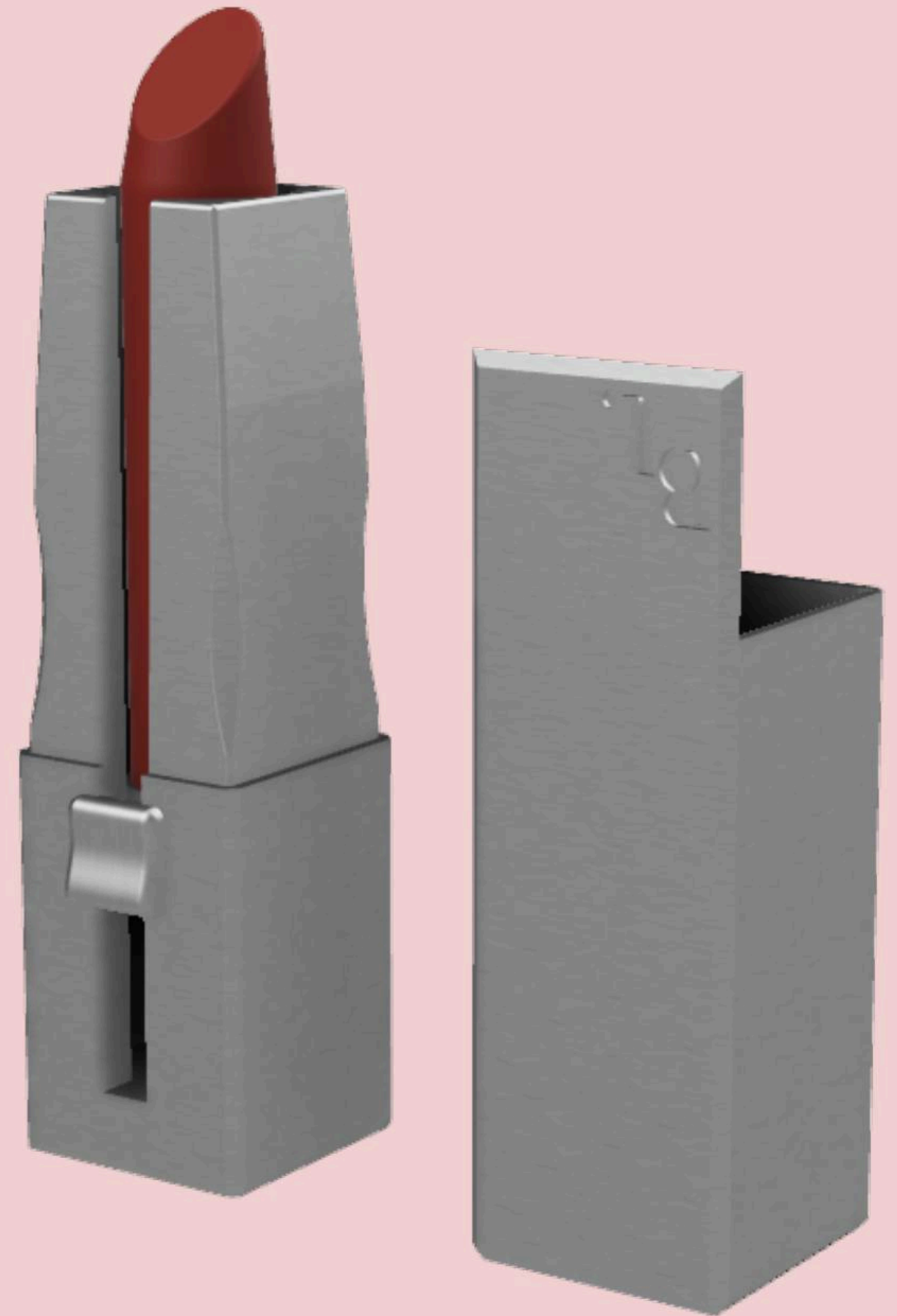


Brushed Lips

Lipstick that Lasts

**SUSTAINABLE DESIGN
ENGINEERING**

Ashley Yang
Gexing Fang
Ruby Kennedy
Sinai Rhodes



PROBLEM OUTLINE

Problems with the current product, user, system

PRODUCT

Nowadays, the vast majority of lipsticks packaging are primarily single-use and not designed for disassembly.

- For instance, a spinning mechanism is typically used for pushing a lipstick out of its container, which is often done by using a pre-manufactured polypropylene tube with spinning mechanism embedded as the container.
- These tubes are covered in a second layer of plastic or coated in metal, usually aluminium, for aesthetic purposes. But because these pieces are frequently **permanently attached to one another**, it is **extremely difficult to separate and retrieve individual parts**.
- Typically, injection-molded plastic parts are used to create the base and cap. Adhesives are used to **permanently join** these parts together which complicates the disassembly.
- Counterweights that exist in some of the designs are difficult to remove.



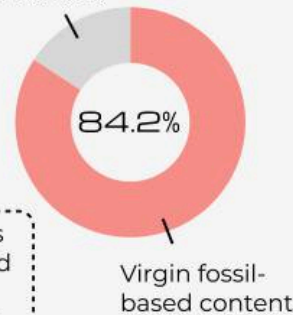
- In order to keep the lipstick standing straight, almost one-third of it was inserted into the tube. Using the parts inside tubes is difficult, and most of the time they are wasted.

The data presented below pertains to the use of plastics in L'Oréal's product packaging, which is typical for well-known cosmetic brands. [1]

Post-consumer recycled content

About **84.2%** of plastics were virgin materials. Only **15.8%** were recycled plastics.

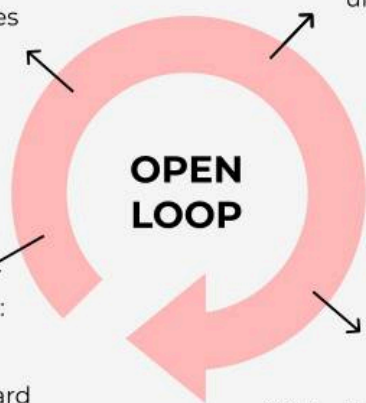
40.7% of L'Oréal's plastics packaging were designed to be **recyclable**. Only **1%** were designed to be **reusable**.



SYSTEM

Distribution:
Because fashion changes rapidly, there is always unsold lipstick in stock that is wasted.

Resources & Production:
The majority of the components were assembled without regard for recycling and were using virgin materials.



Consumption & Use:
Marketing strategies encourage consumers to purchase multiple lipsticks in different colours, even if they are unable to use them all.

Obsolescence & Recovery:

- Small items, particularly plastics, are often challenging to sort and detect in MRFs because of their small size and light weight. This leads to a massive waste of small plastic items, including lipstick packaging.
- Although many lipstick packaging are technically fully recyclable, only a small portion of them were actually recycled. According to the British Beauty Council, only **14%** of cosmetic packaging actually makes it to a recycling plant, only **9%** is **recycled** and the **rest heads directly to landfill**. [2]

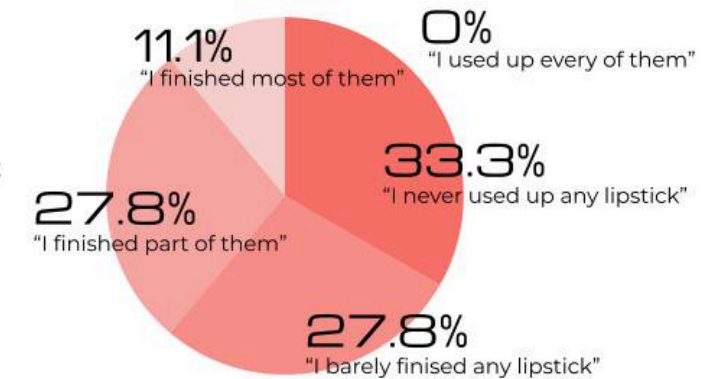
- While there are businesses like Terracycle that recycle cosmetic packaging, they don't offer their services for all brands, products, or locations. Recycling is inconvenient for customers as well due to a lack of incentives and inaccessibility of collection sites.

REFERENCES:

- www.ellenmacarthurfoundation.org. (n.d.). L'Oréal 2021 Global Commitment report on plastic packaging. [online] Available at: <https://www.ellenmacarthurfoundation.org/global-commitment-2021/signatory-reports/ppu/loreal> [Accessed 27 Oct. 2023].
- British Beauty Council (2023). Packaging. [online] The British Beauty Council. Available at: <https://britishbeautycouncil.com/ppbg/packaging/> [Accessed 19 Oct. 2023].

USER - From the survey we conducted

- Main customers are still female.
- 42.1%** of lipstick users said that they had **very unstable timing** when it came to buying new lipsticks. 26.3% of the lipstick users buy new lipsticks every season.
- Half of these lipstick users own more than 20 lipsticks. The highest number gets to 70. On average, these lipstick users own **24.2 lipsticks per person**.
- When asked if they had ever finished a lipstick, **27.8%** of the users replied that they had done so **very seldom**, and **one-third said they had never done so at all**. Just two people used up the majority of their lipsticks, and nobody finished them all.



Problem:

- Buying lipstick can often be an impulsive decision and leaves one with an excessive number of lipsticks that are never worn or unfinished.**

Frequency of purchase



- When consumers are buying lipsticks, **colour is the most significant consideration** (72.2% of users ranked it as the first and nobody gave it a low rank). Following that, **brand and price are ranked second and third**, respectively, by roughly one-third of the users. **Limited edition**, which is typically regarded as a useful marketing strategy, **did not, however, become very well-liked** among the participants.

Problems:

- Users are encouraged to buy a variety of lipsticks in various colours and types to fit different styles and occasions, which also results in excessive lipsticks.**
- The main reason lipsticks aren't used to their full potential is aesthetic obsolescence.**

- Additionally, **users give fewer considerations on sustainability concerns** when purchasing lipsticks. 63% of users have never purchased sustainable products, while 85% of users have never sought recycling programs when disposing of lipsticks.
- Lipsticks priced **between £10 and £20 are preferred by half of users**.
- 66.7% of consumers say they prefer buying lipsticks from counters carrying cosmetic brands**. Nonetheless, **27.8% of users say they prefer to buy lipsticks online**; this could be because the majority of participants are young.

INSIGHTS AND OPPORTUNITIES

Product

- How might we improve the lipstick design to facilitate recycling and disassembly?
- How might we encourage the manufacturers to incorporate more recycled materials into their products?

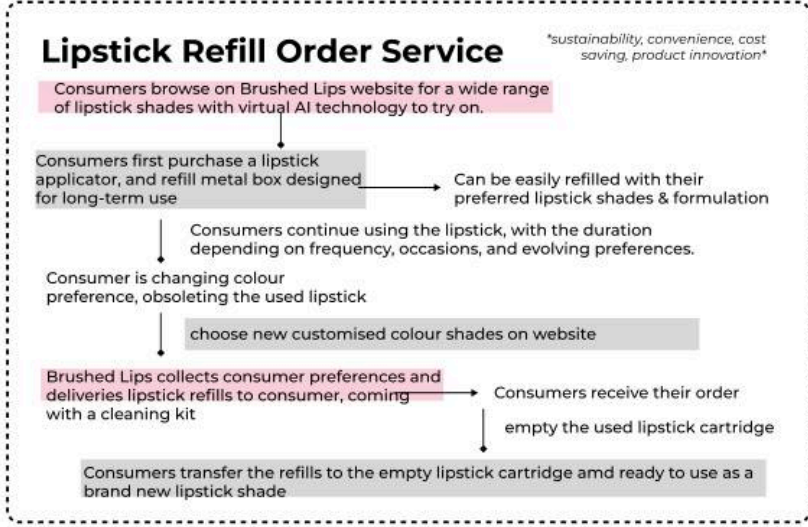
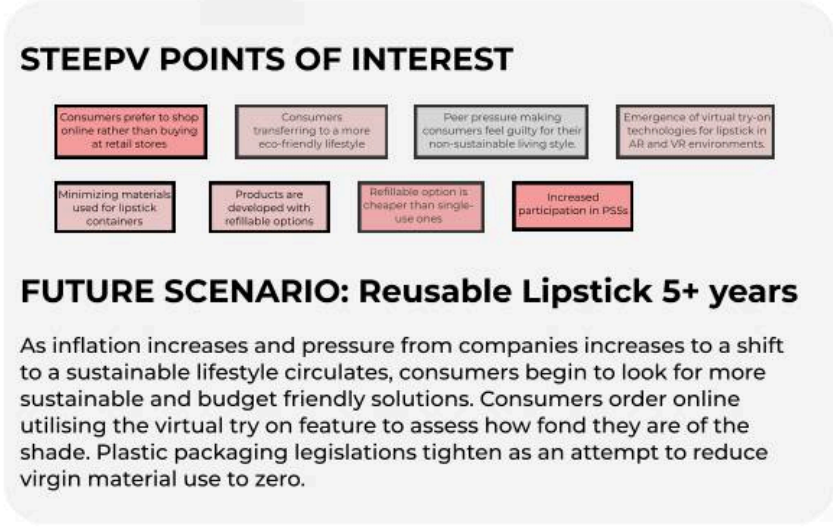
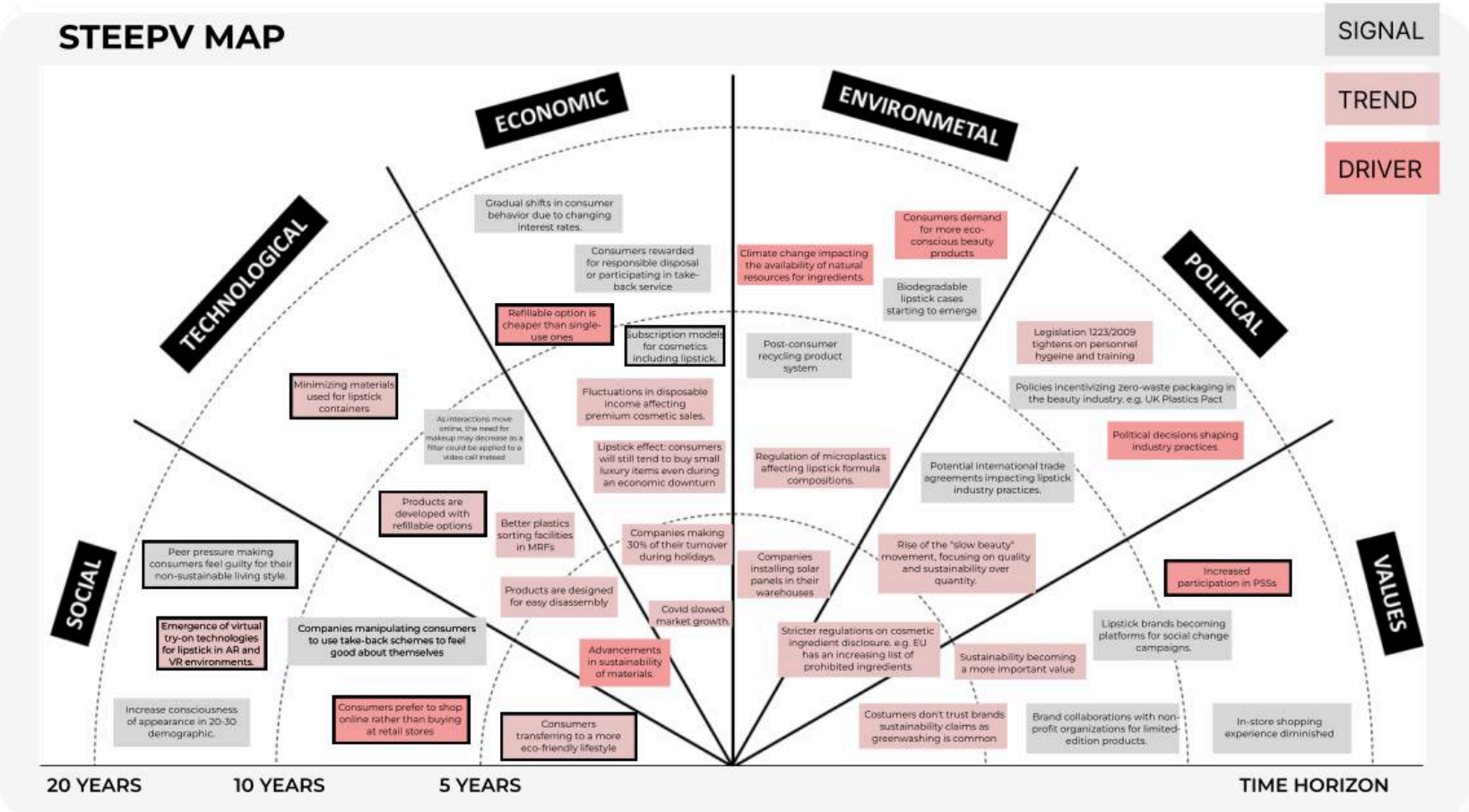
System

- How might we narrow the flow of resources to minimise waste in the system's operation?
- How might we establish a recycling system for lipstick packaging that works well to get the loop more closed?

User

- How might we motivate users to make the most of their lipstick?
- How might we prevent users from purchasing an excessive amount of lipsticks?

FUTURE SCENARIO



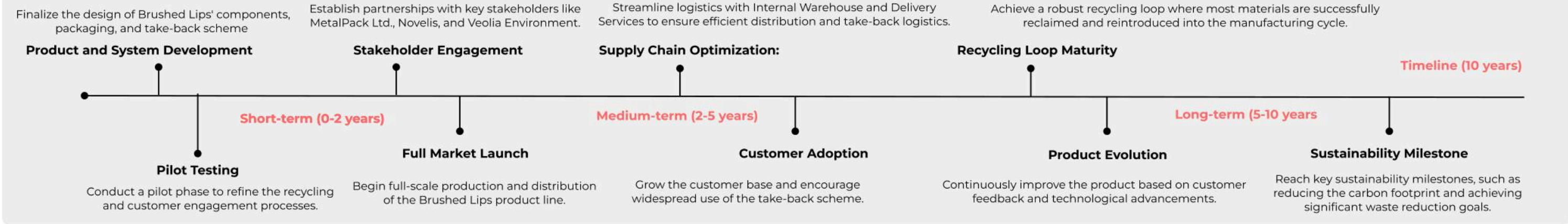
Our ideas

Brushed Lips is pioneering a future where beauty and sustainability are inseparable. Our idea revolves around an elegant, refillable lipstick applicator crafted from 100% recyclable aluminum, designed to last a lifetime. With slots for an applicator, tweezer, and multiple lipstick bullets, along with a biodegradable cleaner, Brushed Lips embodies the principles of conscious consumption and innovative design. Our system includes a take-back scheme, ensuring that every component can be recycled or biodegraded, thereby minimizing environmental impact and fostering a closed-loop system.



- We've taken a risk in our sales model by doing away with retailers and selling lipsticks and refills straight to customers via our website and app. This allows us to better manage our inventory and avoid lipstick waste while simultaneously utilising the internet to make it quicker and easier for customers to buy our products. The frequency of shipping will also increase as a result of this, but overall, the environmental impact of shipping the lipsticks will be minimal.
- As a purely online business, we find it disadvantageous that customers prefer to test the effect in person offline before making a purchase under the current system. For this reason, we have opted to offer customers a virtual try-on experience via AI. If necessary, we can also offer tiny samples to give customers a better try-on experience.

Time Horizon



OUR PRODUCT

Elevate your lipstick experience with Brushed Lips, the Aluminium Refillable Lipstick designed for eco-conscious beauty lovers. The lipstick is complete with an Aluminium case housing applicator and tweezers, biodegradable foam cleaner along with 4 lipstick bullets in a variety of shades. Brushed Lips is the ultimate fusion of conscious beauty and innovative design. At End-of-Life users will use either our take-back scheme where our aluminium is externally recycled and the user will be issued with a new applicator.

Cap
Material: Aluminium
Manufacturing Process: Extrusion
EOL destination: Recycle
Mass: 3g

The Cap provides protection from damage and excessive oxidation to the bullet whilst keeping the slider in place.

Base
Material: Aluminium
Manufacturing Process: Drawn and Milled.
EOL destination: Recycled
Mass: 10g

Drawn and milled with a sleek brushed Aluminium finish. Machined from one piece of Aluminium the base houses the lipstick allowing all components to connect together.

Slider and Holder
Material: Aluminium
Manufacturing Process: Milled
EOL Destination: Reused
Mass: 2g

The slider permits the bullets to move up and down the base to be accessed and the holder clicks into the the slide and is secured onto the lipstick bullet.

Case
Material: Aluminium
Manufacturing Process: Milled and Welded
Mass: 200g

Equipped with an applicator slot, a tweezer slot and 5 bullet slots (4 for lipstick bullets and 1 for the cleaner). The ultimate accessory to complete your lipstick application process. 100% Recyclable as solely made from Aluminium.

Applicator Cleaner
Material: Corn and Bionanopol
Manufacturing Process: Mixing and Baking
Mass: 1g
EOL destination: Biodegradable

Made from biopolymers to biodegrade upon compost and machine to mimic the bullet shape to clean the base before applying a new bullet and the tweezer to prevent contamination between shades.

Tweezers
Material: Aluminium
Manufacturing Process: Die Cut and Induction Welded
EOL destination: Recycled
Mass: 6g

The Tweezers are fully Aluminium with a curved end to allow easy transferral of the bullets between the applicator and case, without contaminating your fingers with the formula or crushing the bullet.

Packaging Design

Our packaging design is fully cardboard allowing for easy disposal. Our cardboard packaging and paper used in wrapping Bullets is FSC certified and 100% recyclable

100% Recyclable Packaging

Serrated strip around paper bullet wrapping for easy removal

Easy-to-use pull out draw packaging

User Experience

Ergonomic Grooves
Designed with your comfort in mind, the ergonomic grooves in the base ensure a secure grip during application. These offer a tactile guide, promoting precise control as during application.

Improved grip

Easier Opening

Better Control

Prevents injuries and cuts

Brushed Aluminium Finish

The allure of Brushed Lips extends beyond function, thanks to its exquisite brushed aluminium finish. The tactile sensation of the brushed surface adds a premium touch to the lipstick, inviting users to indulge in a moment of sophistication every time they reach for this sustainable beauty essential. The brushed finish not only enhances the visual appeal but elevates the overall tactile experience of using Brushed Lips.

Round Edges

The soft rounded edges add a touch of sophistication to the user's beauty ritual. Gentle on the hands and pleasing to the touch, these rounded contours provide a comfortable grip, making every application a luxurious and effortless experience.

Linear pattern

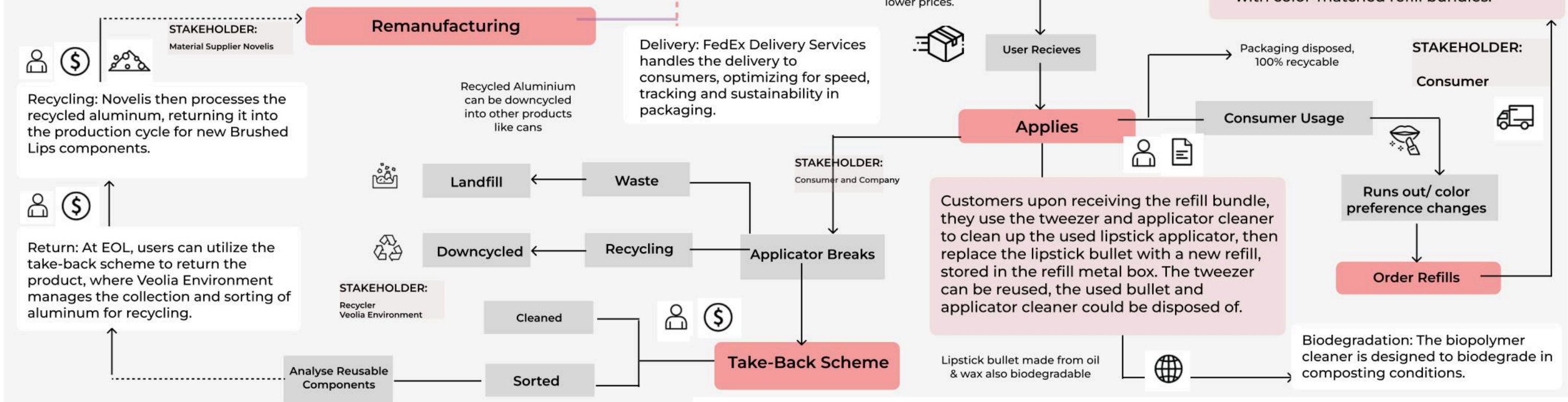
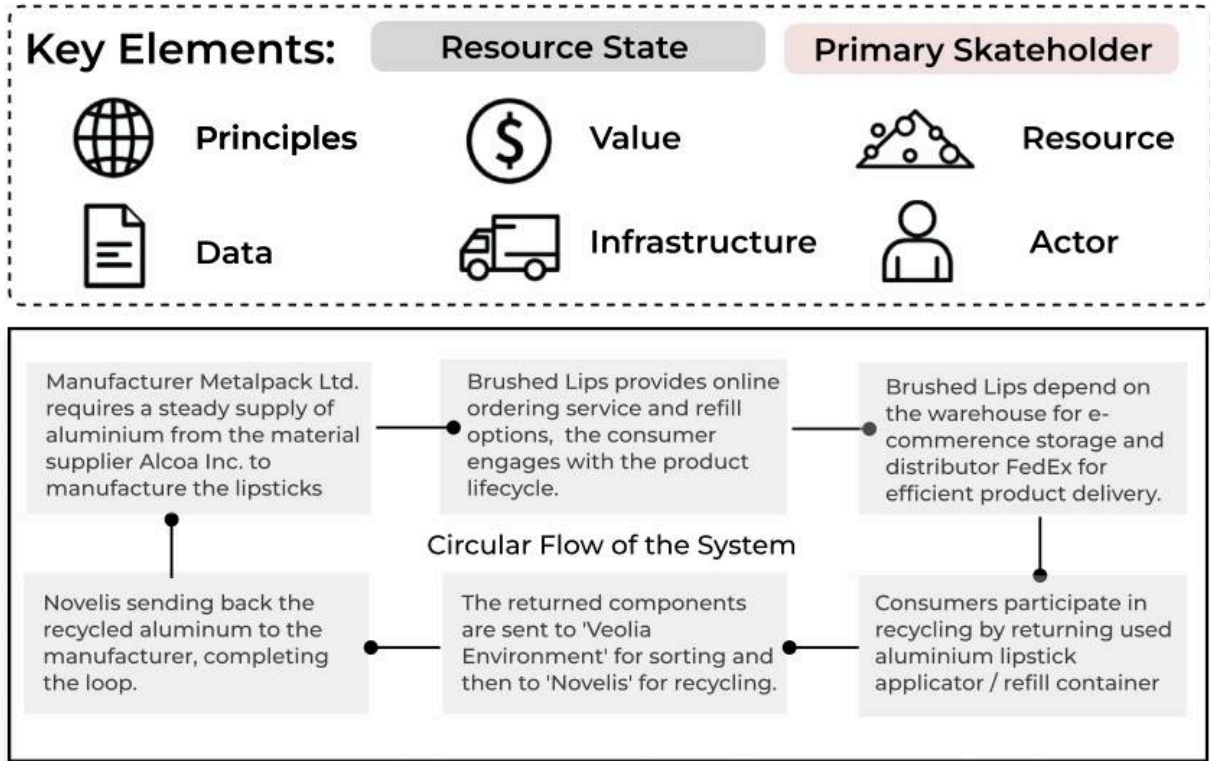
Soft and smooth to touch

SYSTEM DESIGN

Modelling the System holistically by Analysing the Circular Flow or Resources

System Overview:

For Brushed Lips, the circular system design encompasses the entire lifecycle of the product—from creation to end-of-life—ensuring sustainability and a minimized environmental footprint. The system diagram illustrates the processes involved in the path of the product and system, displayed through the six flow mapper system elements and. The primary skateholder involved are encompassed along with resource states labelled.



SOLUTION INNOVATION AND FEASIBILITY

Innovation

Reduces packaging - Having reusable cases reduces the packaging needed for each lipstick.

Reduces transport - The case is the heaviest component of the lipstick. If the consumer keeps the tube and only the refill needs to be shipped this reduces the weight that needs to be transported.

Creates brand loyalty - The refills system makes the consumer more likely to buy again from the company. Consumers are becoming more eco-conscious so are also more likely to be loyal to companies that are more sustainable.

Still allows you to change lipstick shade - one of the biggest problems was consumers wanting to try a different shade of lipstick before they finished the stick. The refill case helps solve this problem.

Feasibility

Problem: Lipstick tubes that are hard to disassemble and recycle with too many materials.

Our Solution: Lipstick tube made only from aluminium and has only a simple sliding mechanism.

Feasibility: Very feasible. It can be designed this way from the start and doesn't rely on unpredictable consumer behaviour in any way.

Problem: Lipstick is only used once and then thrown out.

Our Solution: Refillable lipstick tube and refill system.

Feasibility: Fairly feasible. The system is easy to implement as the bullets of lipstick are already being made and to package them for refill instead of in a case is a minor adjustment. However it is more difficult to guarantee that the customer will buy a refill after the initial purchase. If they do not then the sustainability of the system is compromised. It will lose all of its long lasting and reuse benefits.

Problem: Small metal items are hard to recycle.

Our Solution: Take-back scheme.

Feasibility: Not that feasible. There are already systems in place for recycling aluminium as it is one of the most widely recycled metals. Although with some of these systems the lipstick tube may be too small and get lost. It is also hard to ensure the customer will post the lipstick back on the take-back scheme, despite the incentive of a free replacement case, as it is not super convenient. Also it is not anticipated that the tubes or cases will break frequently. If there were very few items being sent in it may make the scheme obsolete.

ENVIRONMENTAL IMPACT

Eco audit

Assumptions

Manufacturing

All the parts were made with roll forming instead of drawing and die cutting as this was the closest cold forming technique available on the level 2 library

Disposal

All aluminium parts would be disposed of via the take back scheme so they could be recycled

Transport

The shipping route between the raw materials to the factory (Australia to Poland) takes the shipping route through the Suez Canal.

The warehouse is on the outskirts of London.

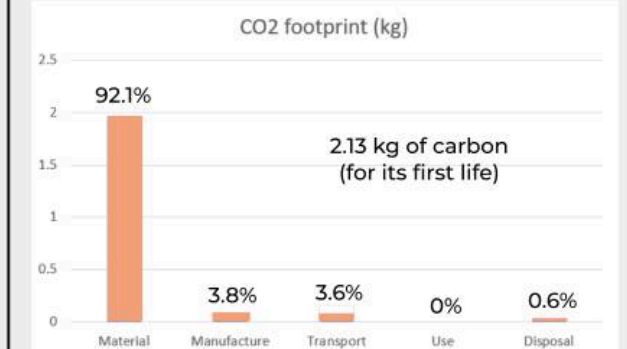
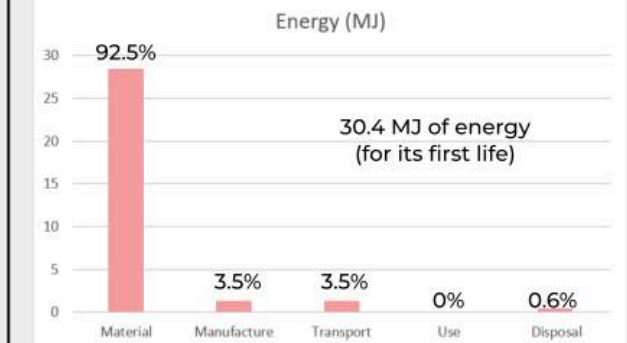
The consumer lives in Central London.

The FedEx truck delivering the lipsticks is 4 ton truck.

Useage

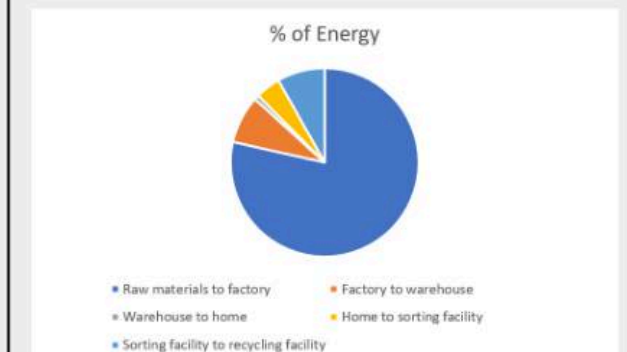
The product will be used for 20 years as it can be refilled and is made of a durable material.

Results



As you can see from the charts the material embodies the most energy and has the highest carbon footprint. However we have chosen this aluminium as it has a high circularity - the material will not lose quality after being recycled multiple times and it is one of the most abundantly recycled metal so there is already infrastructure in place to handle it.

Looking at Transport as it has the second highest embodied energy



The transport of the Raw material to the factory takes the most energy out of all of the transport. This is because it is the longest journey (Australia to Poland). To reduce the energy used here perhaps an aluminium supplier within Europe could be used.

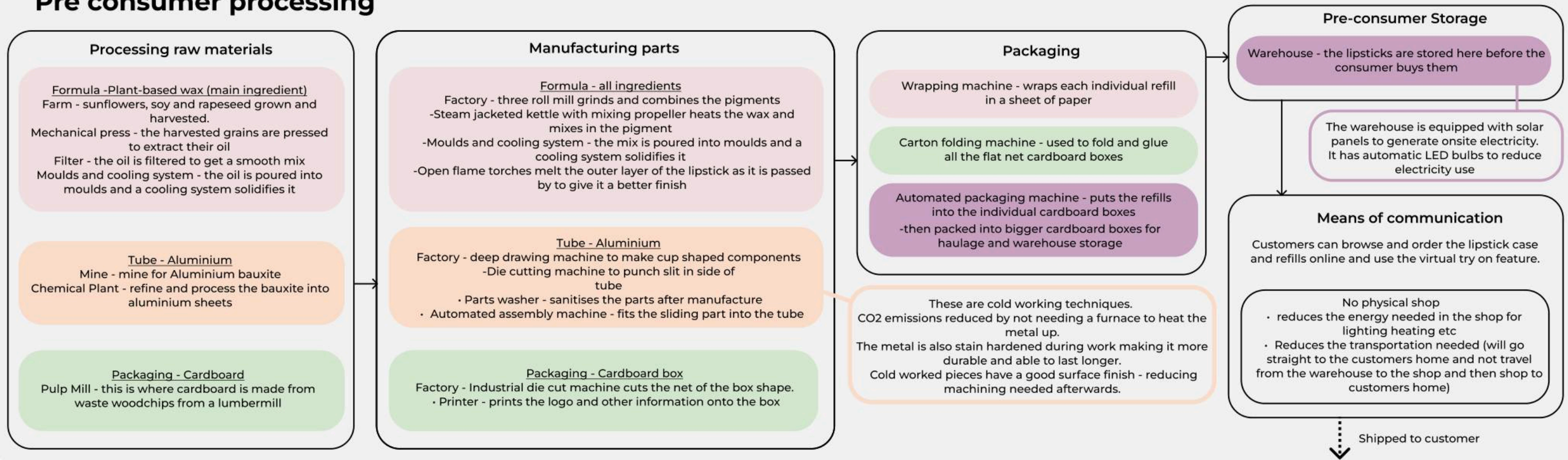
Material Circularity analysis



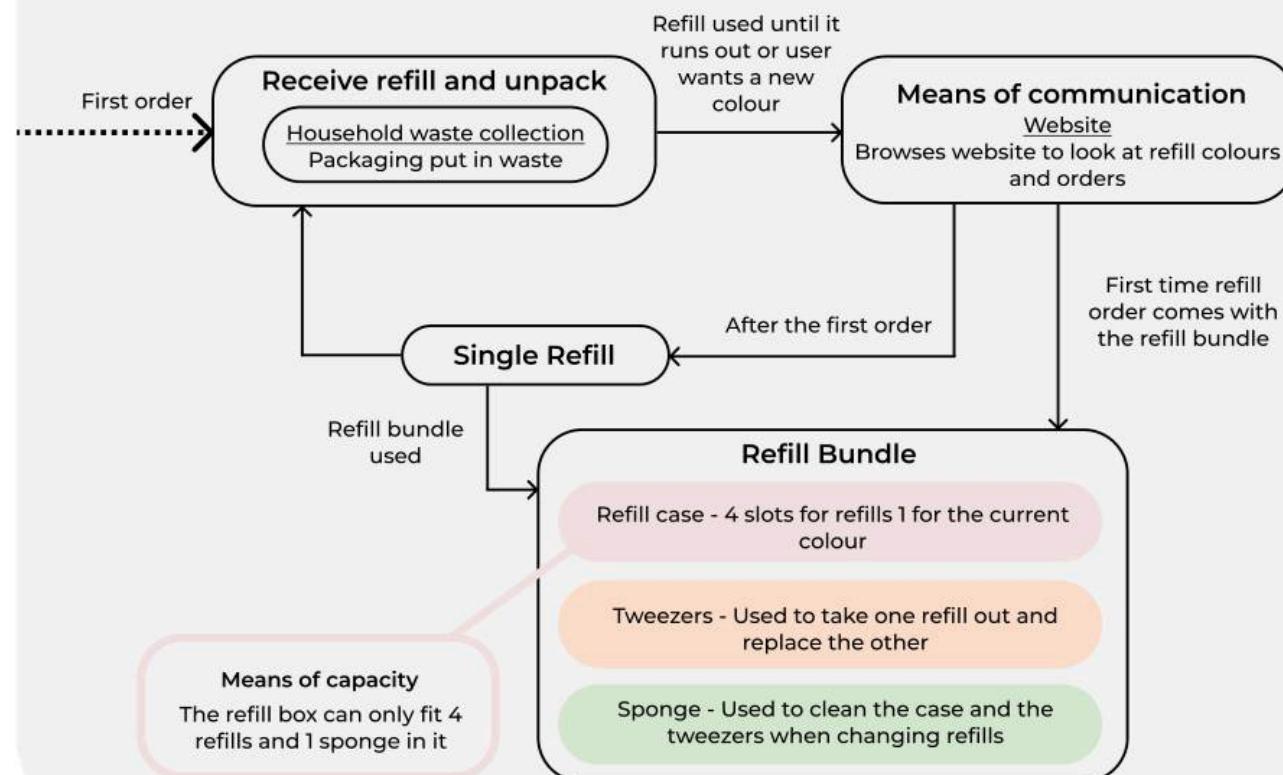
INFASTRUCTURE SYSTEM DESIGN 1

Ruby Kennedy

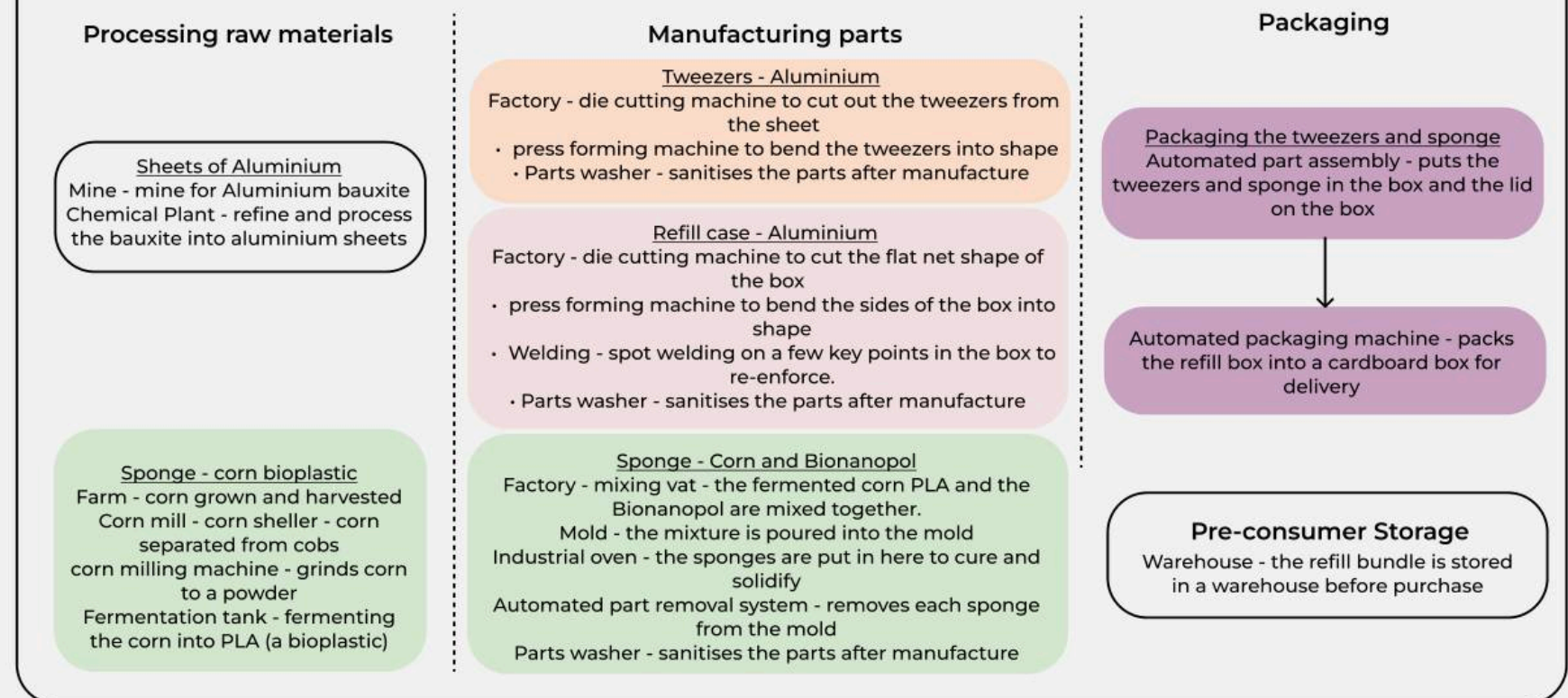
Pre consumer processing



Processing during consumption

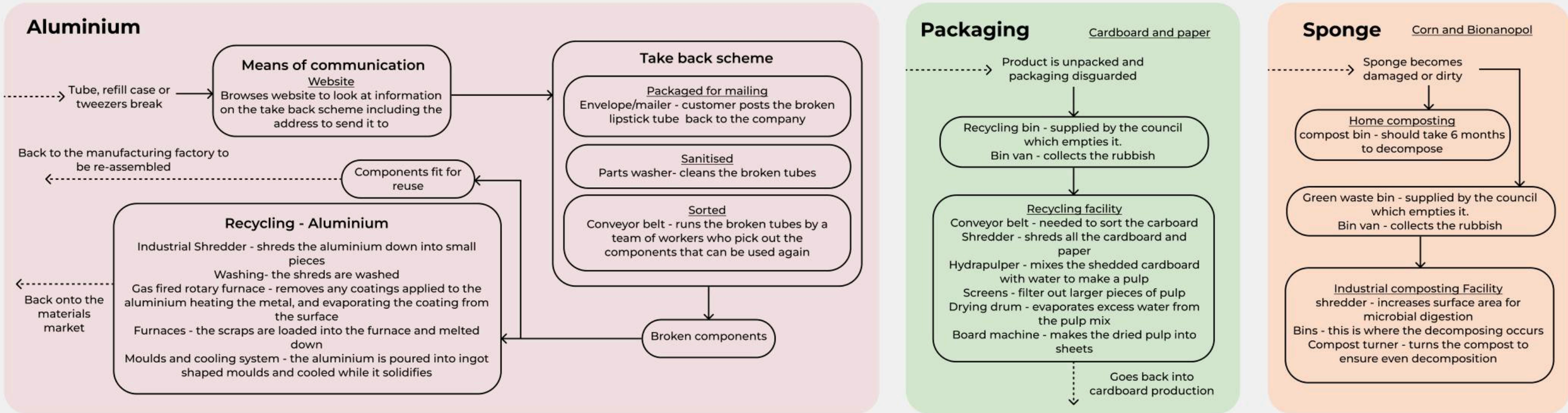


Means of processing pre-consumer - Refill Bundle

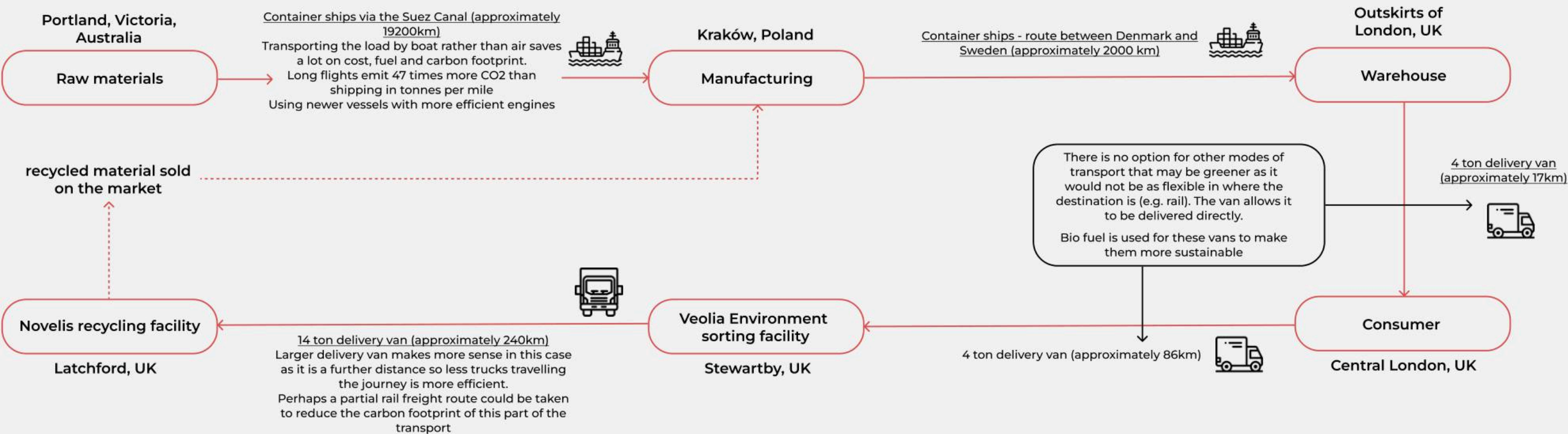


INFASTRUCTURE SYSTEM DESIGN 2

Post consumer processing



Means of transportation - Aluminium



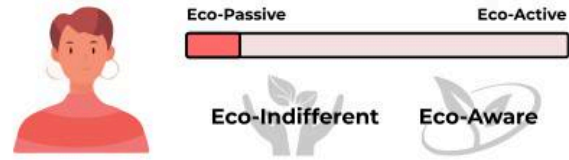
ACTORS SYSTEM DESIGN

Scenario: Emily is going out this weekend and decides to do some online shopping to see if she can find anything to impress her friends. Upon browsing on a cosmetics site she stumbles across an eco-friendly lipstick. She had seen more information about these sustainable makeup alternatives and was in need of a new lipstick.

Sinai Rhodes

Users

Existing User: Wasteful Wendy 18



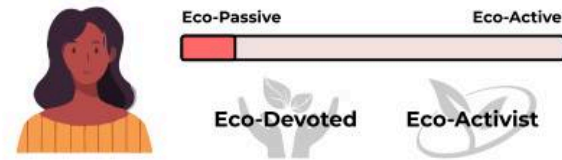
Behaviour and Beliefs

Wendy's approach to lipstick mirrors her overall lack of sustainability consciousness. She doesn't consider eco-friendliness in her decisions and is unconcerned about their environmental impact, prioritises convenience and affordability.

Needs and Goals

Wendy's main goal is to acquire cheap variety of trendy lipsticks to match her daily outfits and stay current with beauty trends. She needs to impress and to be noticed.

Our User: Eco-Conscious Emily 22



Behaviour and Beliefs

Emily is passionate about environmental sustainability and ethical consumption. She avoids single-use plastics and seeks out cruelty-free, eco-friendly products believing every small change can make a big difference.

Needs and Goals

Emily's key needs include lipsticks with recyclable packaging and natural ingredients. She seeks brands with strong sustainability initiatives and products that align with her eco-conscious values.

Stakeholders

Manufacturer: MetalPack Ltd.



Effort: Invest in sustainable practices, explore eco-friendly alternatives, and integrate renewable energy sources in production.

Challenges: Initial cost in eco-friendly technologies and potential disruptions to traditional manufacturing workflows.

Motivation: Gain a positive brand image, increased consumer trust, potential cost savings, and access to environmentally conscious markets.

Distributor: FedEx



Effort: Incorporate sustainable shipping practices, optimize packaging, integrate green transportation methods, and develop efficient logistics, including strategic partnerships.

Motivation: Aligning with corporate sustainability goals, meeting customer demand for eco-friendly shipping, gaining a competitive edge in the market, and achieving long-term cost savings through energy-efficient practices, especially in transporting aluminium products.

Recycler: Novelis



Effort: Implement an efficient recycling system for aluminium and biopolymer-based accessories. Invest in state-of-the-art recycling machinery and establish robust collection networks.

Motivation: The recycling company contributes significantly to a circular economy, reducing the environmental impact of aluminium production, potentially generating revenue from recycling and repurposing materials, and positioning itself as a leader in sustainable waste management.

Consider

Select

Order

Possess

Prepare

Consume

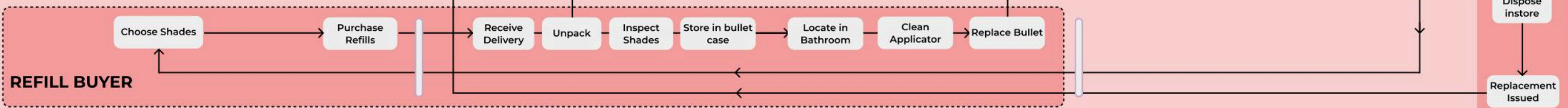
Maintain

Dispose

EXISTING LIPSTICK



FIRST TIME BUYER



REFILL BUYER



"The Virtual try on ensures I've got a shade I'm happy with."

Intrigued

"This delivery is so easy I can't wait until it comes"

Confident

"Changing bullets is so easy and convenient."

Excited

Enthusiastic

"I love this shade, looks just like it did online let me try another."

Impressed

"I can't believe its broken at least I'll get a free replacement."

Hopeful

Content



"There's so many shades to choose from what do I pick"

Overwhelmed

"Let me see how this looks when I get home"

Optimistic

Anxious

Apprehensive

"This shade is nice."

Relieved

"That ran out fast, I guess I'll head to the store to get another."

Disappointed

DATA SYSTEM DESIGN

CLAIMS AND PR



Our Product is 100% Recyclable

80%

We use 80% Recycled Aluminium



We use 100% FSC certified packaging



Our product is 100% disassemblable

Stakeholder Claims

Manufacturer



Our material has the “potential to be recycled infinitely” and “does not lose quality in the recycling process” ¹

Distributor



“Fed-ex branded is packaging is 100% recyclable and composed of 45% recycled content.” ²

Recycler



“Currently 61% of our inputs – is at the core of our innovative, circular business model.” ³

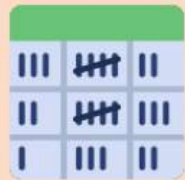
CUSTOMER DATA

Identity



Gather essential customer information, including name, age, location, and gender. This data provides insights into the demographics of your customer base, helping tailor marketing strategies and product offerings to specific segments.

Frequency of Refill Purchases



This data helps identify customer loyalty, preferences for specific shades, and the success of the refillable model. It also guides inventory management to ensure popular refills are consistently available.

Customer Reviews



Analysing reviews helps understand the overall satisfaction of customers, identify popular shades, and address any concerns or areas for improvement. Positive reviews can also be used for marketing and testimonials.

Take-Back Scheme Participation



This data point helps gauge customer engagement with the sustainability initiative. This also provides insights into the durability of the applicator design and customer satisfaction with the replacement process.

REFERENCES:

1. MetalPack EU. (n.d.). Sustainability. [online] Available at: <https://metalpack.eu/sustainability/> [Accessed 27 Oct. 2023].
2. FedEx Business Insights. (n.d.). What Makes FedEx Packaging Sustainable? [online] Available at: <https://fedexbusinessinsights.com/what-makes-fedex-packaging-sustainable/> [Accessed 27 Oct. 2023].
3. Novelis. (n.d.). Sustainability. [online] Available at: <https://www.novelis.com/sustainability/> [Accessed 27 Oct. 2023].

RESOURCE INFO

Current Solution

A combination of non-recyclable plastics and metal can be permanently bonded together to make disposable lipsticks.

Lipsticks are sealed with sellotape which will end up in landfill and are packaged using single use plastics.

When lipstick runs out users leave remaining lipstick in the applicator unable to fully use the lipstick

Quantity Data

Every Brushed Lips applicator is etched with an Identification number not only to personalise the product and make each applicator unique to the user but to make an easy identifier as to the product quantity manufactured relative to those sold.

Our Solution

We use 100% Aluminium in our refillable lipstick making it easy to disassemble and recycle.

Our lipstick has a tight seal limiting oxidation and comes in 100% corrugated recyclable cardboard.

Our design permits full use of our lipstick without compromising material and a cleaner to eliminate residue.



INSTRUCTIONS

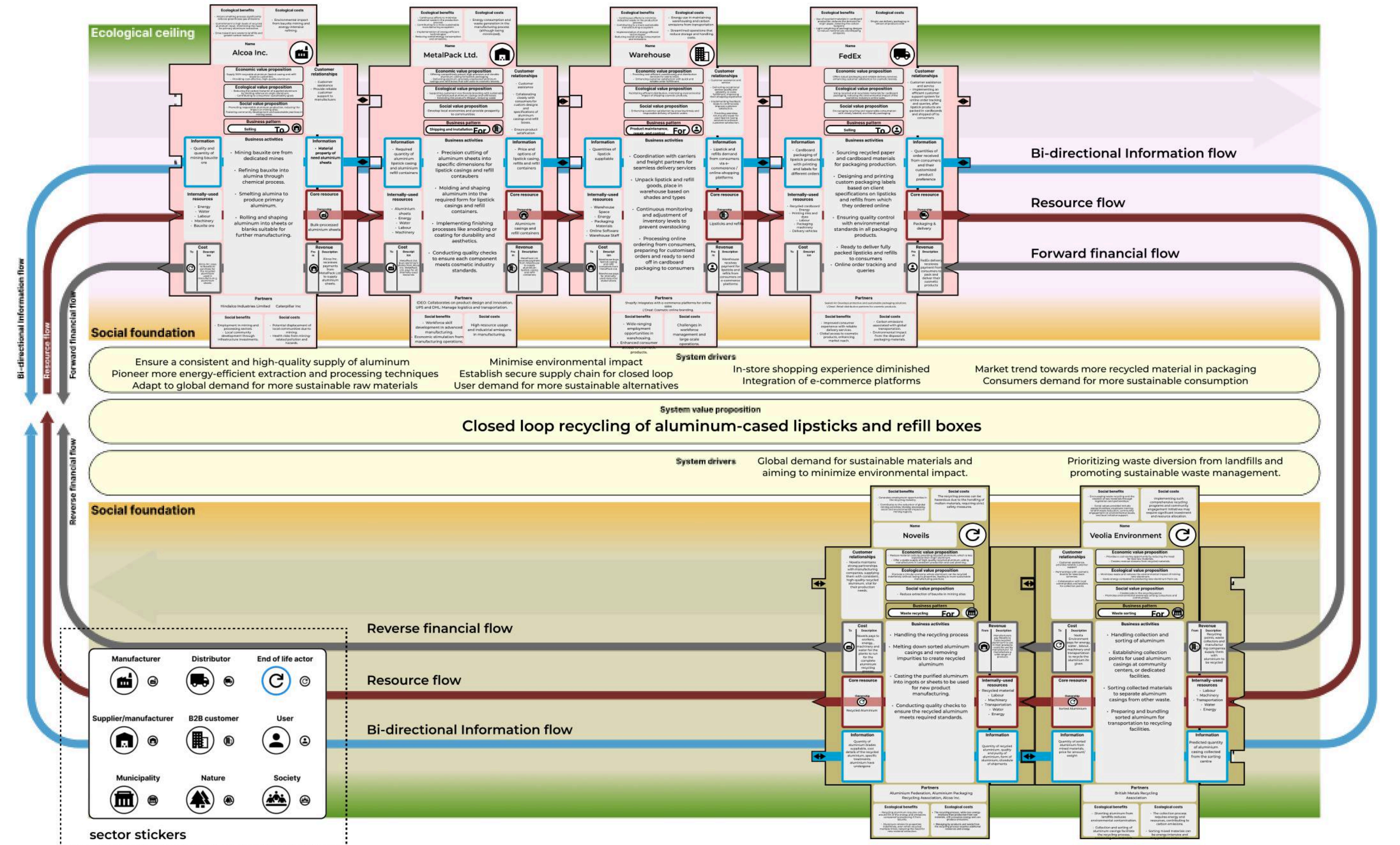
There is the option for a series of processes carried out by the user which they may need guidance. The infographic below shows the details refilling the applicator.



VALUE SYSTEM CIRCULAR DESIGN

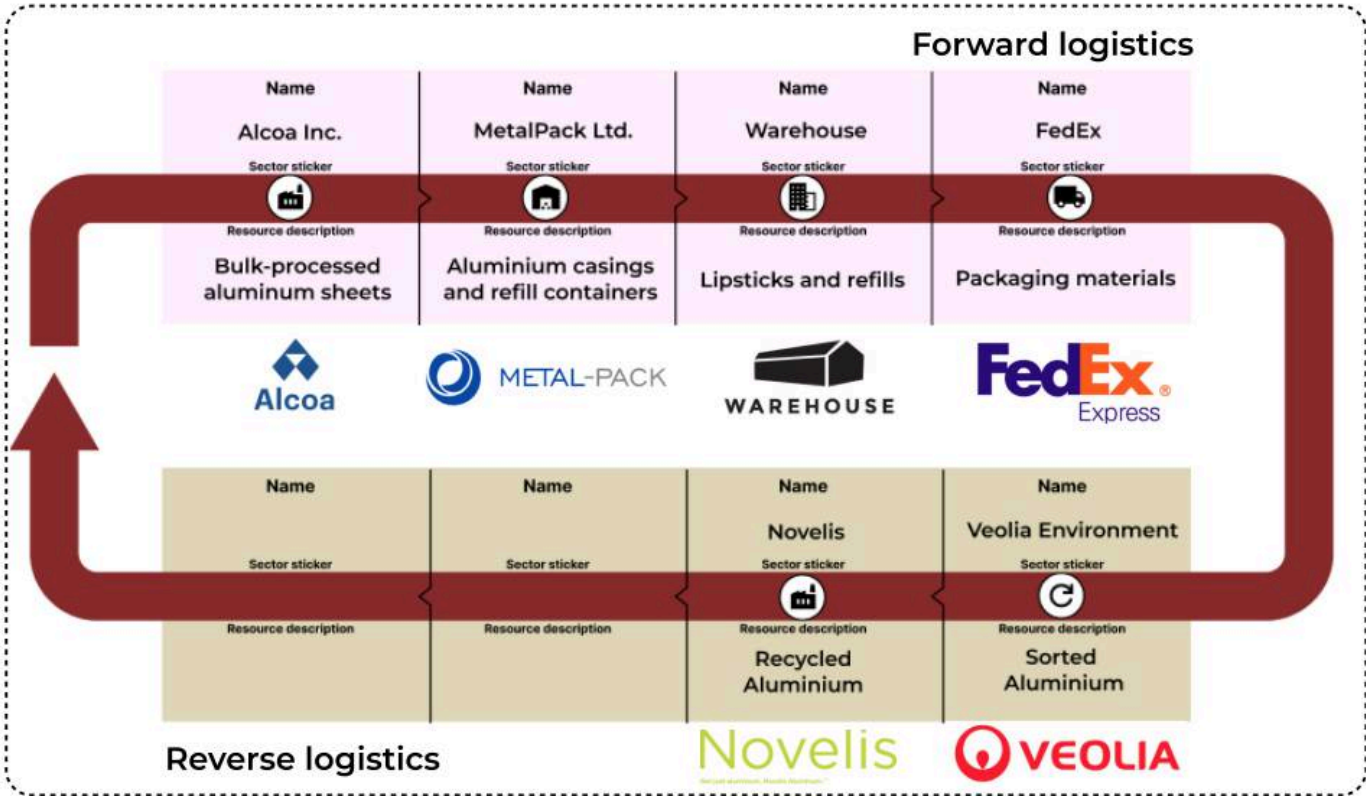
Ashley Yang

Circular Ecosystem Business Model Canvas



VALUE SYSTEM OVERVIEW

System resource flow diagram



The business model canvas

Actors	Key activities	Key resources	Value proposition	Customer relationship	Channels
Alcoa Inc.	Bauxite mining, aluminum refining and processing.	Bauxite ore, processing facilities, skilled labour.	High-quality aluminum supply with a focus on sustainability.	Builds long-term relationships with manufacturers by providing consistent, high-quality aluminum	Direct sales to manufacturers, industry partnerships.
MetalPack Ltd.	Manufacturing aluminum casings and boxes.	Aluminum sheets, manufacturing equipment, skilled labour	Precision-engineered aluminum casings for lipstick industries.	B2B relationships with clients through collaborative product development, customization	Direct sales to manufacturers, cosmetic industry
Warehouse	Storage, inventory management, order fulfillment.	Warehouse facilities, logistics infrastructure, workforce.	Streamlined warehousing and fulfillment services for online consumer orders.	E-commerce integration, customer service.	Online retail platforms, direct B2B services.
FedEx	Packaging, logistics, delivery services.	Delivery trucks, logistics network, workforce.	Reliable, efficient packaging+delivery services tailored for e-commerce businesses.	Delivery tracking, customer support.	Direct B2B services, online delivery platform
Veolia Environment	Waste collection, sorting, and recycling management.	Collection trucks, sorting facilities, workforce.	Efficient waste management and recycling services.	B2B sales, customer service.	Local and regional waste management
Novelis	Aluminum recycling, quality assurance, material supplier	Recycled aluminum, recycling facilities.	Sustainable, high-quality recycled aluminum.	Long-term manufacturing contracts, collaborative development.	Direct sales to manufacturers, industry collaborations.
Cost Structure Mining operations, processing and refining costs. Manufacturing and operational costs. Warehousing operations, labour, inventory management. Transportation, packaging material cost, delivery operations, staff. Waste collection and sorting operations. Recycling processes, innovation and development.			Revenue Stream Sales of processed aluminum. Sales of aluminum casings and boxes. Fulfillment service fees, storage fees. Delivery and customer service fees. Fees from waste management services, sales of recyclables. Sales of recycled aluminum.		

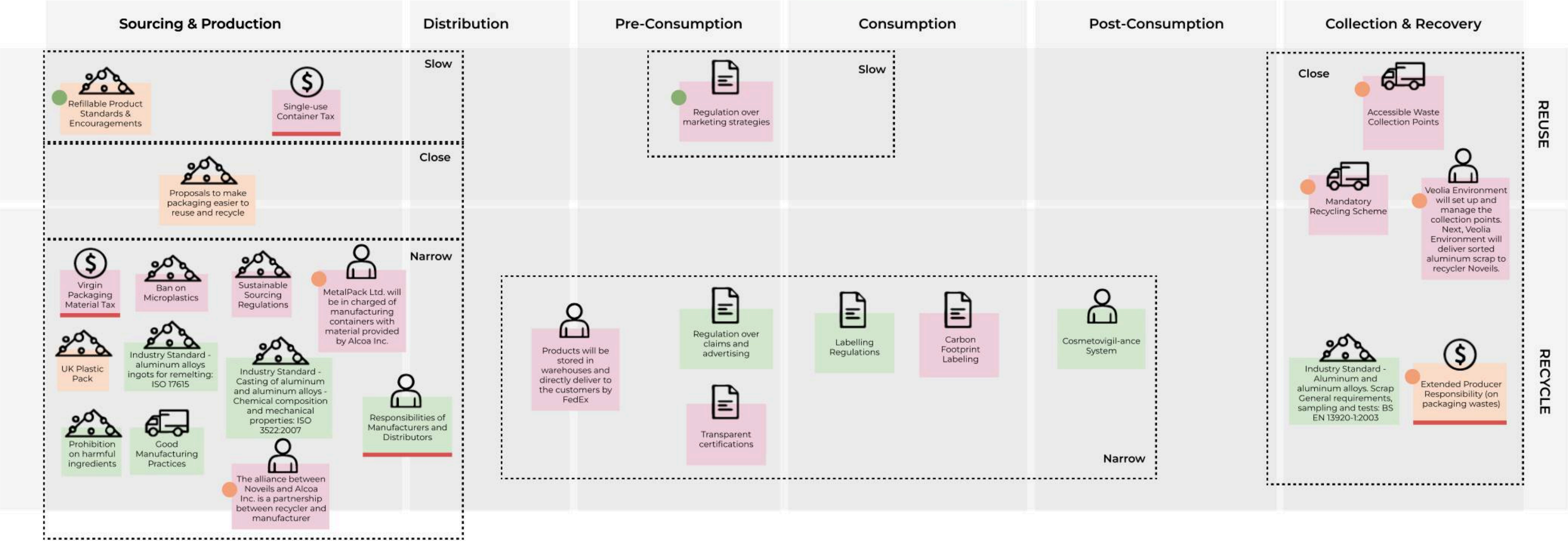
Analyse the canvas - Identifying risks and opportunities

Category	Revenue/benefits: What are the positive impacts of the system?	Cost: What are the negative impacts of the system?	Cost mitigation: How can the negative impacts be mitigated?
Finances	Alcoa Inc. provides aluminum to manufacturers, generating revenue. MetalPack Ltd. sells aluminum casings to companies, creating value. Internal Warehouse streamlines distribution for businesses. FedEx ensures timely deliveries, increasing customer satisfaction. Veolia facilitates recycling and environmental conservation. Novelis supplies recycled aluminum, reducing primary production costs.	Alcoa Inc. and MetalPack Ltd. face high operational costs. Warehouse Services deals with logistics expenses. Veolia's costs stem from waste collection and sorting infrastructure.. Novelis has energy costs for recycling.	Novelis optimises recycling processes, looks for lowest aluminium price and make supply aggrements. Alcoa Inc. undergoes efficient manufacturing, and logistics operations. Veolia Environment can improve waste collection efficiency, lower processing cost and lost aluminium by educating consumers.
Ecology	Recycling aluminum for casings uses less energy, reducing the carbon footprint. Novelis's process of recycling aluminum requires only around 5% of the energy and emissions compared to producing it from bauxite. Aluminium can be recycled indefinitely without losing its original property. Maintain a high level of recycled-metal input. Warehouse efficient logistics network minimizes indirect transportation emissions.	Not all aluminium containers are recycled, contributing to land and sea pollution. Not all aluminium containers are recycled, contributing to value loss, emissions from truck transportation of aluminium. The extraction and processing of aluminum can lead to environmental degradation, and the manufacturing process can generate waste and emissions.	Brand could educate consumers to dispose of aluminium casings + refills appropriately Advancements in recycling technology. Use of renewable energy in manufacturing and electric vehicles for transportation of aluminium. Veolia can focus on maximizing recycling rates to reduce waste.
Society	Alcoa and MetalPack Ltd. create jobs and support local economies. Warehouse's services offer convenience and support economic activity. Overall job creation across the recycling, manufacturing, and logistics sectors. Community development through environmental initiatives.	Potential social issues include community displacement by mining activities and the labor-intensive nature of waste sorting and processing jobs.	Alcoa can invest in local community programs to offset mining impacts. Veolia and Warehouse can ensure fair labor practices and invest in workforce development. MetalPack Ltd. can offer employee training programs to ensure job retention and satisfaction.

PRINCIPLE SYSTEM DESIGN

Gexing Fang

• Prohibition on harmful ingredients	Bans the use of specific toxic or environmentally damaging substances in lipstick formulations. (EU, ECHA)	DR	• Ban on Microplastics	Prohibits the use of microplastics in lipsticks to reduce environmental pollution.	DR
• Plastic Packaging Tax	Imposes a tax on plastic packaging with less than 30% recycled plastic to incentivize the use of recycled plastic. (UK)	EI	• Refillable Product Standards & Encouragements	Sets standards and encourages or mandates the use of refillable containers for lipsticks. (EU)	DR, EII
• UK Plastic Pack	A commitment in the UK to achieve specific targets for reducing plastic waste in the packaging industry.	VI	• Carbon Footprint Labeling	Requires labeling that shows the carbon footprint of lipstick production and disposal.	DR
• Good Manufacturing Practices (GMP)	All cosmetics products must be manufactured in accordance with the harmonised standards laid out in GMP (EU, CE)	DR	• Sustainable Sourcing Regulations	Mandates the use of sustainably sourced ingredients in lipsticks.	DR
• Labelling Regulations	Requires clear labeling of ingredients, sourcing, and environmental impact on lipstick packaging. (EU, CE)	DR	• Virgin Packaging Material Tax	Imposes a tax on new, non-recycled packaging materials to encourage the use of recycled content.	EI
• Responsibilities of Manufacturers and Distributors	Mandates that these entities ensure product safety, regulatory compliance, and ethical distribution practices. (EU, CE)	DR	• Transparent certifications	Develops certification systems to transparently indicate sustainable and ethical practices in lipstick production.	DR, EII
• Regulation over claims and advertising	Governs the accuracy and truthfulness of environmental and health claims in lipstick marketing.	DR, VI	• Accessible Waste Collection Points	Establishes convenient locations for the collection of cosmetic waste, including lipstick containers.	DR
• Cosmetovigilance System	A system for monitoring and reporting adverse reactions or effects from cosmetic products, including lipsticks. (EU, CE)	DR	• Single-use Container Tax	Levies a tax on single-use lipstick containers to discourage their use and promote sustainable alternatives.	DR
• Extended Producer Responsibility (on packaging wastes)	Obligates manufacturers to manage and reduce the environmental impact of their packaging waste. (EU)	DR	• Governments & Associations Investments	Encourages government and industry association investments in research and development for sustainable lipstick packaging and production technologies.	EI
• Proposals to make packaging easier to reuse and recycle	Introduces regulations to encourage the design of lipstick packaging that is more easily recyclable or reusable. (EU)	DR	• Regulation over marketing that promotes over-consumption	This policy aims to regulate and limit advertising strategies that encourage excessive purchasing and usage of lipsticks, thereby fostering more sustainable consumer habits and reducing waste.	DR, EII
• Mandatory Recycling Scheme	Requires the recycling of lipstick packaging, potentially including consumer return systems.	DR	• Industry Standards:	Aluminum alloys ingots for remelting: ISO 17615; Casting of aluminum and aluminum alloys - Chemical composition and mechanical properties: ISO 3522:2007; Aluminum and aluminum alloys. Scrap General requirements, sampling and tests: BS EN 13920-1:2003; The alliance between Novelis and Alcoa Inc. is a partnership between recycler and manufacturer; MetalPack Ltd. will be in charged of manufacturing containers with material provided by Alcoa Inc; Products will be stored in warehouses and directly deliver to the customers by FedEx; Veolia Environment will set up and manage the collection points. Next, Veolia Environment will deliver sorted aluminum scrap to recycler Novelis.	DR
			• Agreements:		



All Phases, All Strategies, All Loops

Governments & Associations Investments

Key

Policy Environment: Restrictive None Supportive

System Element: Value Actor Data Infrastructure Resource

Resource Life Cycle: S&P, D, Pre-C, C, Post-C, C&R

Circular Business Model: Repair Rental Resale Remake

Circular Economy Loop: Recycle, Reuse

Maturity: Idea In development Established

Circular Economy Strategy: Narrow, Slow, Close

EXISTING PRINCIPLE SYSTEM & NEW ELEMENTS

Gexing Fang

Idea In development Established

EXISTING SYSTEM

ANAYLSIS

NEW ELEMENTS (PART OF)

• Prohibition On Harmful Ingredients

 Bans the use of specific toxic or environmentally damaging substances in lipstick formulations. (EU, ECHA)
Category: DR

• Plastic Packaging Tax

 Imposes a tax on plastic packaging with less than 30% recycled plastic to incentivize the use of recycled plastic. (UK)
Category: EI

• Good Manufacturing Practices (GMP)

 All cosmetics products must be manufactured in accordance with the harmonised standards laid out in GMP (EU, CE)
Category: DR

• Labelling Regulations

 Requires clear labeling of ingredients, sourcing, and environmental impact on lipstick packaging. (EU, CE)
Category: DR

• Regulation over claims and advertising

 Governs the accuracy and truthfulness of environmental and health claims in lipstick marketing.
Category: DR, VI

• Cosmetovigilance System

 A system for monitoring and reporting adverse reactions or effects from cosmetic products, including lipsticks. (EU, CE)
Category: DR

• Responsibilities of Manufacturers and Distributors

 Mandates that these entities ensure product safety, regulatory compliance, and ethical distribution practices. (EU, CE)
Category: DR

• Industry Standards (Plastic)

 Industry Standard - acrylonitrile-butadiene-styrene (ABS) molding and extrusion: ISO 19062-1:2015
Industry Standard - Plastics - Polyethylene (PE) moulding and extrusion materials: ISO 1872:2007
Industry Standard - quality management systems: ISO 9001:2015
Industry Standard - Guidelines for the recovery and recycling of plastics waste: ISO 15270:2008
Industry Standard - Plastics - Generic identification and marking of plastics products: ISO 11469:2016

- EU has a growing list of prohibited ingredients that contains 2495 unique substances/entries as of 12 December 2023.
- By mandating manufacturers to use and label ingredients from sustainable sources, we hope to go one step further in this regard.

Replaced
Under the current system, a tax is levied to discourage packaging made entirely of virgin plastic. Nonetheless, the current system continues to consume a significant amount of plastics, including some that are difficult to recycle and may build up in the environment. Therefore, we are implementing a few new policies to stop this issue from getting worse.

- Current labelling regulations do not prioritise increasing consumers' awareness of environmental protection. A carbon footprint could be included on the label to help customers understand how their actions affect the environment.

- The current beauty industry wastes a lot of stuff, especially when it comes to lipstick. Many lipsticks aren't being used to their full potential.
- By creating a number of guidelines, we hope to improve upon the current policy and motivate users to switch to more sustainable purchasing patterns.

- The cosmetics industry uses a tremendous amount of plastic for packaging these days, but the majority of plastics are not recycled (even though it is perfectly possible for them to be recycled) because of the tiny size of these plastic pieces and the absence of a specialised recycling system.
- Therefore, we designed a recycling system centred on the more recyclable aluminium alloy that we used in place of the plastics in lipstick containers, guaranteeing a high rate of recycling of the container material during system operation.

Replaced

• Sustainable Sourcing Regulations

 Mandates the use of sustainably sourced ingredients in lipsticks.
Category: DR

• Ban On Microplastics In Lipstick Ingredients

 Prohibits the use of microplastics in lipsticks to reduce environmental pollution.
Category: DR

• Virgin Packaging Material Tax

 Imposes a tax on new, non-recycled packaging materials to encourage the use of recycled content.
Category: EI

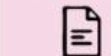
• UK Plastic Pack

 A commitment in the UK to achieve specific targets for reducing plastic waste in the packaging industry.
Category: VI

• Governments & Associations Investments

 Encourages government and industry association investments in research and development for sustainable lipstick packaging and production technologies.
Category: EI

• Carbon Footprint Labeling

 Requires labeling that shows the carbon footprint of lipstick production and disposal.
Category: DR, EII

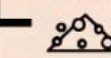
• Regulation over marketing that promotes over-consumption

 This policy aims to regulate and limit advertising strategies that encourage excessive purchasing and usage of lipsticks, thereby fostering more sustainable consumer habits and reducing waste.
Category: DR, EII

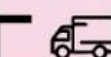
• Transparent certifications

 Develops certification systems to transparently indicate sustainable and ethical practices in lipstick production.
Category: DR, EII

• Refillable Product Standards & Encouragements

 Sets standards and encourages or mandates the use of refillable containers for lipsticks. (EU)
Category: DR, EII

• Mandatory Recycling Scheme

 Requires the recycling of lipstick packaging, potentially including consumer return systems.
Category: DR

• Extended Producer Responsibility (on packaging wastes)

 Obligates manufacturers to manage and reduce the environmental impact of their packaging waste. (EU)
Category: DR

• Single-use Lipstick Container Tax

 Levies a tax on single-use lipstick containers to discourage their use and promote sustainable alternatives.
Category: DR

• Industry Standards (Aluminum)

 Industry Standard - aluminum alloys ingots for remelting: ISO 17615
Industry Standard - Casting of aluminium and aluminium alloys - Chemical composition and mechanical properties: ISO 3522:2007
Industry Standard - Aluminium and aluminium alloys. Scrap General requirements, sampling and tests: BS EN 13920-1:2003